

Saurav G Mishra

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Education

Gopalan College of Engineering and Management (VTU)

Bangalore, India

B.E. in Information Science & Engineering (2nd Year) | Expected Graduation: 2029 | Year 1 CGPA: 7.72 / 10

Technical Skills

Languages: Python, JavaScript, C, C++, SQL, HTML, CSS

Systems & Low-Level: Linux Syscalls, POSIX, C++20, QEMU, Process Management, Signals, Socket ioclt

Web & Backend: Node.js, Firebase (Auth, Cloud Firestore), SQLite, Flask, REST APIs

Automation & Tools: Git, GitHub, Bash, n8n, Zapier, Docker (Basics), Activepieces

AI-Assisted Development: Cursor, Antigravity, Claude Code, Codex

Experience & Internships

AI Research Intern — Sunny Side Up

May 2025 – Jun 2025

In-office, Bangalore

- Researched and tested 50+ AI tools and APIs across image generation, media, and productivity to support team adoption.
- Prototyped Zapier automation workflows for internal team reminders and evaluated Photoshop ExtendScript for layout adaptation.
- Documented tool features, limitations, and reproducible setup workflows for internal engineering reference.

Projects

FreeDot Linux — Minimal C++ Userspace & Init System [\[GitHub\]](#)

Personal Project

C++20, Linux Syscalls, POSIX, QEMU, Bash

- Built a custom PID 1 init system replacing standard init daemons on a custom Linux LTS kernel, handling orphan process reaping (waitpid), signal handling, and service supervision.
- Configured network interfaces directly via raw socket ioctls (SIOCSIFFLAGS) and packaged a static rootfs into an initramfs tested in QEMU under a strict 600MB memory ceiling.

StudyForge — Gamified Study & Task Tracker [\[GitHub\]](#)

Personal Project

JavaScript, HTML5, CSS3, Firebase Authentication, Cloud Firestore

- Built a client-side productivity app that turns study tasks into a game with XP progression, levels, and daily streak tracking.
- Integrated Firebase Authentication and Cloud Firestore for real-time task persistence and profile synchronization.

FoodShare — Community Food-Sharing Platform [\[GitHub\]](#)

Personal Project

JavaScript, Node.js, Firebase Firestore, PWA

- Developed a Progressive Web App for local communities to list and share surplus food to minimize waste, with real-time listing feeds and responsive mobile layout.

Assignmate — Peer-to-Peer Student Tutoring Marketplace [\[GitHub\]](#)

Personal Project

Python, Flask, SQLite, HTML5, CSS3

- Implemented a student tutoring platform with role-based auth, task bounties, and relational SQLite schema for course assistance.

CB-watch — Gesture-Controlled Rubik's Cube Timer [\[GitHub\]](#)

Personal Project

Python, OpenCV, Google MediaPipe

- Built a touchless timer script using OpenCV and MediaPipe hand landmark tracking to detect open-palm gestures for starting, stopping, and resetting without keyboard interaction.

VI — Personal Voice Assistant Prototype [\[GitHub\]](#)

In Progress

Python, SpeechRecognition, Groq API, File I/O

- Created a desktop voice assistant script experimenting with microphone speech input, Groq LLM queries, and saving memory notes to local Markdown files.